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Oxbow Restorations for Topeka Shiner (Notropis topeka) Recovery: Defining Success

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ABSTRACT.—Within the Great Plains region of North America, oxbow wetlands were lost as streams were channelized and straightened to allow for agricultural use of the landscape. Topeka Shiner, which rely on these oxbow wetlands, have declined in abundance, currently occupy 20% of their historical geographic range, and are now an endangered species. A large emphasis has been placed on restoring oxbow wetlands to recover Topeka Shiner populations. As the number of oxbow restorations has increased, so has the number of monitoring projects focused on these habitats. Currently, conservation practitioners rely on the presence of Topeka Shiner to determine the success of an oxbow restoration. Given the rarity of the species, this method is likely ineffective. Within this study we attempt to create a model that can be used to determine the success of an oxbow restoration that will benefit the Topeka Shiner without relying on the presence of the species in the wetland. We used multiple fish community metrics and modelling strategies to determine the best model for predicting Topeka Shiner CPUE and presence/absence within oxbows. We determined that the top models were ineffective at predicting Topeka Shiner CPUE and presence/absence. Our findings illustrate the random nature by which fish communities become established within oxbows, and the island-like nature of oxbows in terms of how fish communities may become established and change over time. These findings indicate that the success of oxbow restoration for Topeka Shiner needs to be defined at a larger scale than for individual oxbows. Furthermore, our findings corroborate the inability of previous studies to determine reproducible associations of Topeka Shiner with various fish community metrics.

INTRODUCTION

Human land use and resulting declines in aquatic ecosystem habitat abundance and quality have led to declines in freshwater fish abundance and species diversity (Jelks *et al.*, 2008; Gido, Dodds, and Eberle, 2010; Hoagstrom, Brooks, and Davenport, 2011; Nislow *et al.*, 2011). Based on rates of fish population declines and habitat loss, Ricciardi and Rasmussen (1999) conclude the extinction rate of North American freshwater fishes will be approximately 2.4% per decade over the next century. Other estimates range from 130 to 855 times the natural background rate of extinction (Burkhead, 2012; Tedesco *et al.*, 2013). To reverse these trends, an increasing emphasis is being placed on the restoration of aquatic habitats.

Oxbow wetlands (hereafter oxbows) naturally form when a river meander is cut off as the channel migrates within its floodplain (Wohlman and Leopold, 1957). However, as rivers were channelized for agricultural land use, floodplain habitats were transitioned to

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cropland. Channelization inhibits the natural migration of the river channel and creation of new oxbows. Furthermore, over time, existing oxbows naturally fill in with sediment and eventually cease to exist. Loss of existing ecosystems, combined with lack of production of new wetlands, results in ever-declining numbers of oxbows in agricultural landscapes.

The Topeka Shiner (*Notropis topeka*) is a minnow (family Leuciscidae; Gidmark and Simons, 2014; Tan and Armbruster, 2018) that is endemic to the Great Plains region of the United States. Historically, this species was common within the states of Kansas, Nebraska, Missouri, Iowa, South Dakota, and Minnesota, where it most often occurred in low-order streams and associated off-channel habitats (Moore, 1957). Currently, the species occupies approximately 20% of its historical geographic range (Tabor, 1998; Wall and Berry, 2004; Mammoliti, 2014). The Topeka Shiner has experienced widespread population declines and range contractions due to loss and degradation of critical habitat (including oxbow wetlands), which led to the listing of Topeka Shiner as federally endangered in 1998 (Tabor, 1998).

Previous studies have determined that oxbows are critical habitat for Topeka Shiner (Minckley and Cross, 1959; Pflieger, 1997; Wall and Berry, 2004; Loan-Wilsey *et al.,* 2005; Bakevich, Pierce and Quist, 2013). This species spawns in areas of slow-moving water (Mammoliti, 2014), and adults and juveniles tend to be more abundant within oxbows than other habitats (Simpson, 2018). Therefore, loss of oxbow habitat has likely contributed to the decline of Topeka Shiner populations.

Oxbows have been restored in Iowa, Minnesota, and South Dakota both to restore the ecosystem services they provide, and to promote the recovery of the Topeka Shiner. The goal of these restoration projects in terms of Topeka Shiner recovery is to increase the amount and quality of critical Topeka Shiner habitat across the landscape, which will hopefully lead to increases in Topeka Shiner populations as a result. Furthermore, many oxbows have been restored within areas of the historical range of Topeka Shiner but in places where Topeka Shiner have not been present in recent years. These restorations follow a "if you build it, they will come" strategy in the hopes that restoring habitat in areas where Topeka Shiner once occurred will lead to the return of the species to the area. Early signs have been documented that point towards this strategy as being successful (Osterhaus *et al.*, 2021).

During restoration oxbows are restored by excavating topsoil down to the former stream bed depth, which results in a reestablished groundwater connection, and greater water depths and volume (Kenney, 2013; Simpson *et al.*, 2019). In addition, a narrow connection to the adjacent stream on the downstream end of the oxbow is excavated in areas where reconnection is unlikely during a typical 2-to-3-y flood event (Schilling *et al.*, 2019). This connection allows fishes to move between the oxbow and associated stream during flood events (Fischer *et al.*, 2018). In Iowa, the rate at which oxbows are being restored has increased from 1.2 restorations per year from 2001–2005 to 12.8 restorations per year from 2016–2020 (Osterhaus *et. al.,* 2021).

Acceleration of oxbow restoration has led to increased monitoring of these ecosystems. Monitoring has focused on the value of restored oxbows as habitats for Topeka Shiner and has sought to identify abiotic and biotic variables that are correlated with Topeka Shiner presence or absence (Bakevich *et al.*, 2013; Fischer *et al.*, 2018; Simpson *et al.*, 2019). The overarching goal of these monitoring projects is to generate information needed to restore and manage oxbows in ways that maximize their value as habitat for Topeka Shiner.

To this point, conservation practitioners have relied on the simple presence or absence of Topeka Shiner to assess the quality or success of an oxbow restoration. However, given the rarity of Topeka Shiner within its geographic range, relying on presence/absence data to

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assess restoration success may be an ineffective approach. Furthermore, many of these oxbows are restored in areas where Topeka Shiner are thought to be extirpated. The absence of the species from these sites may not be an indication of a poor-quality restoration, but rather may be due to the fact that the species simply has not yet had time or opportunity to re-colonize these sites. Consequently, relying on the presence or absence of the species within these oxbows would likely be ineffective. Therefore, a method is needed to assess the quality of oxbow restorations in terms of the potential benefit to Topeka Shiner populations, without relying on the presence of the species within the oxbow to determine success.

A multi-metric index, which could predict the presence or absence of Topeka Shiner within an oxbow, and if present, the abundance of Topeka Shiner, would allow conservation practitioners to quantify the quality or success of an oxbow restoration more effectively. Although previous studies have determined specific oxbow fish community variables associated with the presence or absence of Topeka Shiner, these associations have not been replicated between studies (Bakevich *et al.*, 2013; Fischer *et al.*, 2018; Simpson *et al.*, 2019), and datasets used to analyze the associations were relatively small. It is possible that analysis of a larger, more comprehensive dataset, would allow for associations between Topeka Shiner and specific fish community variables to become more apparent.

Consequently, we conducted a study with the objective of developing an Oxbow Quality Index (OQI) that is based on models that are predictive of the presence or absence of Topeka Shiner within a restored oxbow, and when present, the abundance (*i.e.*, population size or density). The OQI will be used to generate scores for oxbow sites that are representative of the level of success of the restoration in terms of the potential presence/absence or abundance of Topeka Shiner with high quality restorations receiving high scores and low-quality restorations receiving low scores. Based on this criterion we consider oxbows with a high predicted probability of presence of Topeka Shiner, or a high predicted abundance, to be high-quality oxbows. Conversely, low-quality oxbows are those which have a low predicted probability of presence of Topeka Shiner. Unlike previous studies that analyzed data collected for the specific study, our study analyzed data compiled from multiple Topeka Shiner monitoring projects (Bakevich, 2012; Bybel, 2018; Simpson, 2018; Osterhaus *et al.,* In Review). Creation of the OQI would allow for conservation practitioners to assess effectiveness of an oxbow restoration without relying on documented presence of Topeka Shiner as the sole indicator of restoration success.

METHODS

Data for this study come from a database comprised primarily of past Topeka Shiner monitoring projects conducted in Iowa and Minnesota (Fig. 1) by our research group (Bakevich, 2012; Bybel, 2018; Simpson, 2018; Osterhaus *et al.,* In Review). Seining was used to collect fish community data in all of these studies. Seining implementation was consistent between studies and used a single pass of a bag seine (10.7 m $\times$ 1.8 m, 0.6 cm mesh or 16.8 m $\times$ 1.8 m, 0.6 cm mesh depending on oxbow width) along the entire length of the oxbow. For oxbows $<$16 m in average width, the 10.7 m wide seine was used. For oxbows $>$16 m in average width, the 16.8 m wide seine was used. Simpson *et al.* (2019) found a single seine pass (rather than multiple passes) was sufficient for surveying the fish community of an oxbow. A single seine pass is more desirable compared to multiple seine passes given the reduced effort, and decreased fish stress and mortality (Bakevich *et al.*, 2013).

Data are available for the species of fishes collected at each oxbow (presence/absence), the abundance of each species, and for several abiotic and biotic habitat variables. Abiotic and biotic variables were measured immediately prior to fish sampling and methods were

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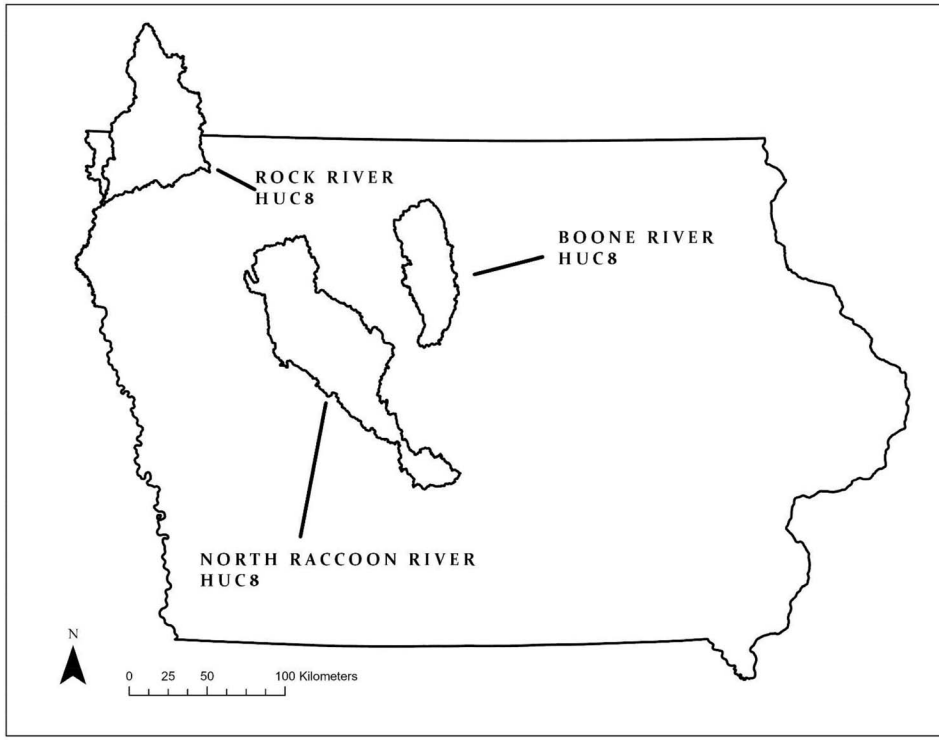


FIG. 1.—Locations of basins in which studies were conducted in Iowa and Minnesota, U.S.A.

consistent between studies (Bakevich, 2012; Bybel, 2018; Simpson, 2018; Osterhaus *et al.,* In Review). All sampling methods were approved by the Iowa State University Institutional Animal Care and Use Committee. Oxbow fish communities are likely structured by a multitude of biotic and abiotic factors both at the site and watershed level (Fig. 2). Given

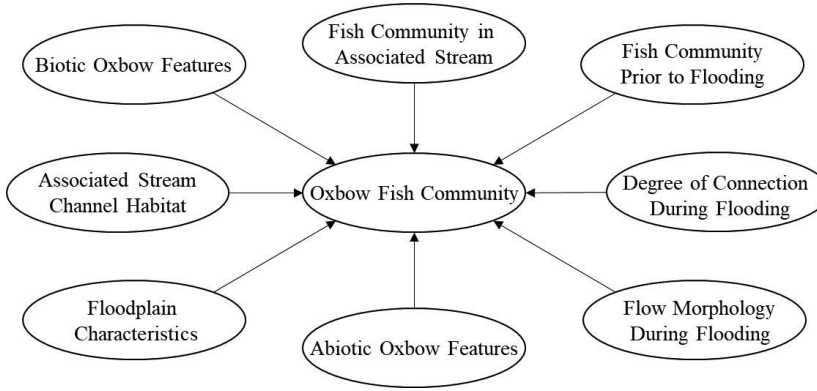


FIG. 2.—Diagram of potential variables driving fish community composition, and in turn, the presence/absence and abundance of Topeka Shiner within oxbows

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that it is likely not feasible to measure and then model each of these factors, we chose to model Topeka Shiner presence, absence, and abundance directly from the fish community data. This modelling strategy operates with the assumption that fish communities where Topeka Shiner are present are similar to each other, and different than those where they are absent as a result of the driving biotic and abiotic factors which have been found by other studies (Bakevich, 2012; Bakevich *et al.*, 2013; Gerkin and Paukert, 2013; Simpson, 2018; Simpson *et al.*, 2019). Using this method we attempted to identify a set of fish community metrics that were significantly correlated with Topeka Shiner presence/absence or abundance. Additionally, an OQI that can be calculated solely from fish community data would allow for expedited implementation of the OQI in the field by restoration practitioners (*e.g.* Iowa Soybean Association, The Nature Conservancy, and United States Fish and Wildlife Service).

To supplement the existing data and generate an additional set of data for model analysis, two seasons of field work were completed during the summers of 2019 and 2020. Fish community data from 137 oxbows within the Boone (42.753, –94.006), North Raccoon (42.328, –94.871) and Rock River (43.458, –95.826) watersheds (Fig. 3) were included in the combined database and analyzed with methods described below. Data from the additional set of 72 oxbow sites sampled within the same region during 2019 and 2020 were used as an additional test dataset to analyze model performance. Sampling methods for sites included within the test dataset were similar to those of previous studies (Bakevich, 2012; Bybel, 2018; Simpson, 2018).

FISH COMMUNITY METRICS AND DATA ANALYSIS

A set of 70 candidate metrics based upon fish community data were calculated for each of the 137 sites included in the database (Table 1). Candidate metrics were taken from previous studies examining Topeka Shiner correlations in oxbows (Bakevich, 2012; Bybel, 2018; Simpson, 2018) and from other fish-based habitat quality indices (Lyons *et al.*, 2001; Drake and Pereira, 2002). Redundancy among candidate metrics was analyzed via Spearman's rank correlation and metrics with strong correlations with other metrics ($r < 0.7$) were considered to be redundant. Because including multiple predictor variables that are highly correlated with each other produces poor model results, when redundant metrics were identified, we used only one such metric in additional analysis. The metric selected was the one for which data are easiest to collect or interpret or was more ecologically meaningful in terms of Topeka Shiner presence/absence or abundance. For instance, the CentrarchidCPUE and GreenSunfish-CPUE metrics (*see* Table 1 for metric definitions) were found to be highly correlated, consequently, GreenSunfishCPUE was retained for further analysis as only one species is included in this metric and is therefore easier to calculate. The remaining candidate metrics were then ranked by their strength of correlation with Topeka Shiner catch per unit effort (CPUE; individuals/100m^2) to generate a final candidate metric set for Topeka Shiner CPUE analyses (Table 1). Remaining candidate metrics were also ranked by their strength of correlation with Topeka Shiner presence/absence to generate a final candidate metric set for Topeka Shiner presence/absence analyses (Table 1). The reduction of the candidate metric list from 70 potential metrics to 15 for each metric set (CPUE and Presence/Absence) was necessary to allow for efficient model selection implementation. Metrics were grouped into three types, CPUE (CPUE for single species or additive CPUE for metrics with multiple species), richness (species richness for species included within the metric), and composition (% of the total catch of the single species or multiple species included in the metric).

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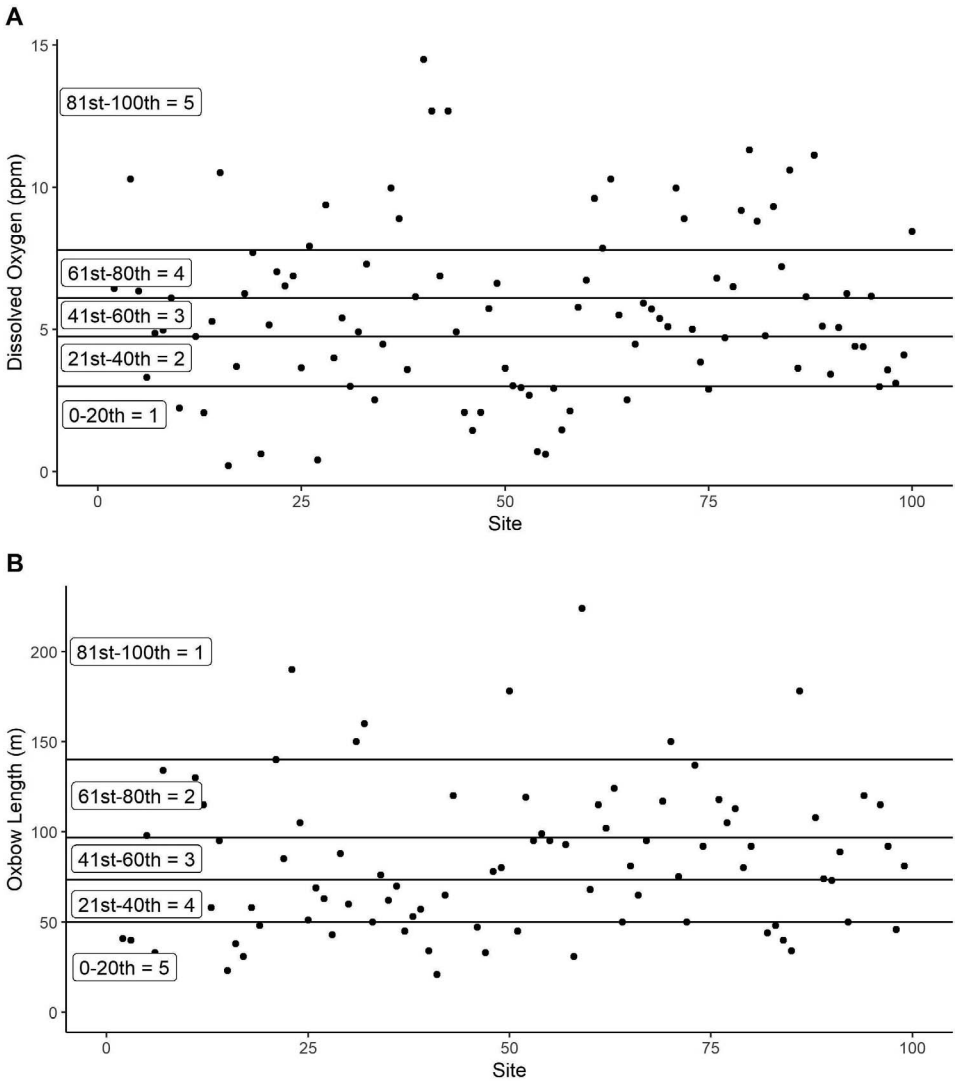


FIG. 3.—Examples of site scoring procedure for (**A**) a positively correlated environmental variable (dissolved oxygen) and (**B**) a negatively correlated environmental variable (oxbow length)

Modelling Topeka Shiner CPUE and presence/absence using fish community metrics.—Modeling was conducted to determine top models for predicting Topeka Shiner CPUE and presence/absence from the fish community metrics. An additive negative binomial linear model that allows for overdispersion was fit to the Topeka Shiner CPUE data (using the glm.nb function in package MASS), while an additive logistic regression was fit to Topeka Shiner presence/absence (using the glm function). Predictor variables were each of the three metric types from the appropriate candidate metric set. Null models were generated for each of the three metric types for Topeka Shiner CPUE and presence/absence. Top candidate models for

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TABLE 1.—Candidate metrics evaluated for inclusion in the OQI. Composition metrics were calculated as the percent composition of the fish community as the species included within the metric. CPUE metrics were calculated as the CPUE of the species included in the metric at the oxbow. Richness metrics were calculated as the number of species at the oxbow included in the metric

Metric name	Metric definition	Metric type	Included in Topeka Shiner CPUE candidate metric set?	Included in Topeka Shiner presence/absence candidate metric set?
CompGrOr	Green Sunfish + Orangespotted Sunfish	Composition	Yes	Yes
CompInvore	Invertivores	Composition	Yes	Yes
CompSens	Sensitive Species	Composition	No	Yes
CompTolerant	Tolerant Species	Composition	No	Yes
CompCent	Centrarchids	Composition	No	No
CompGrOrFa	Green Sunfish + Orangespotted Sunfish + Fathead Minnow	Composition	No	No
CompHerb	Herbivores	Composition	No	No
CompLent	Lentic Species	Composition	Yes	No
CompOmni	Omnivores	Composition	No	No
CompOr	Orangespotted Sunfish	Composition	No	No
CompPisc	Piscivores	Composition	No	No
CompPiscCrCh	Piscivores + Creek Chub	Composition	No	No
CompPiscGS	Piscivores + Green Sunfish	Composition	No	No
CompPiscGSCrCh	Piscivores + Green Sunfish + Creek Chub	Composition	No	No
CompSGCN	Species of Greatest Conservation Need	Composition	No	No
CompTop3	Three most abundant species at the site	Composition	No	No
CompTopCarn	Top Carnivores	Composition	No	No
BluegillCPUE	Bluegill	CPUE	No	Yes
CommonCarpCPUE	Common Carp	CPUE	No	Yes
CrappieCPUE	White Crappie + Black Crappie	CPUE	No	Yes
LargemouthBassCPUE	Largemouth Bass	CPUE	Yes	Yes
PiscivoresCrChCPUE	Piscivores + Creek Chub	CPUE	No	Yes
SGCNCPUE	Species of Greatest Conservation Need	CPUE	No	Yes
SpotfinShinerCPUE	Spotfin Shiner	CPUE	No	Yes
TopCarnCPUE	Top Carnivores	CPUE	No	Yes
YellowBullheadCPUE	Yellow Bullhead	CPUE	No	Yes
BassCPUE	Largemouth Bass + Smallmouth Bass	CPUE	No	No
BenthicCPUE	Benthic Invertivores	CPUE	No	No
BigmouthShinerCPUE	Bigmouth Shiner	CPUE	No	No
BlackBullheadCPUE	Black Bullhead	CPUE	Yes	No
BluntnoseMinnowCPUE	Bluntnose Minnow	CPUE	No	No
BrassyMinnowCPUE	Brassy Minnow	CPUE	No	No
BrookSticklebackCPUE	Brook Stickleback	CPUE	No	No
CentrarchidCPUE	Centrarchids	CPUE	No	No

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TABLE 1.—Continued

Metric name	Metric definition	Metric type	Included in Topeka Shiner CPUE candidate metric set?	Included in Topeka Shiner presence/absence candidate metric set?
CommonShinerCPUE	Common Shiner	CPUE	Yes	No
CreekChubCPUE	Creek Chub	CPUE	Yes	No
FatheadCPUE	Fathead Minnow	CPUE	Yes	No
GizzardShadCPUE	Gizzard Shad	CPUE	No	No
GreenSunfishCPUE	Green Sunfish	CPUE	Yes	No
GrOrCPUE	Green Sunfish + Orangespotted Sunfish	CPUE	No	No
GrOrFaCPUE	Green Sunfish + Orangespotted Sunfish + Fathead Minnow	CPUE	No	No
HerbCPUE	Herbivores	CPUE	No	No
InvertCPUE	Invertivores	CPUE	No	No
JohnnyDarterCPUE	Johnny Darter	CPUE	No	No
LenticCPUE	Lentic Species	CPUE	Yes	No
OmnivoreCPUE	Omnivores	CPUE	No	No
OrangespottedSunfish CPUE	Orangespotted Sunfish	CPUE	Yes	No
PiscivoresCPUE	Piscivores	CPUE	No	No
PiscivoresGrCPUE	Piscivores + Green Sunfish	CPUE	No	No
PiscivoresGrCrChCPUE	Piscivores + Green Sunfish + Creek Chub	CPUE	No	No
SandShinerCPUE	Sand Shiner	CPUE	No	No
SensitiveCPUE	Sensitive Species	CPUE	No	No
TolerantCPUE	Tolerant Species	CPUE	No	No
Top3CPUE	Three most abundant species at the site	CPUE	No	No
TotalCPUE	Total	CPUE	Yes	No
WhiteCrappieCPUE	White Crappie	CPUE	No	No
WhiteSuckerCPUE	White Sucker	CPUE	Yes	No
ShannonIndex	Shannon Diversity Index	Diversity	No	No
CentRich	Centrarchids	Richness	No	Yes
PiscivoreRich	Piscivores	Richness	Yes	Yes
BenvertRich	Benthic Invertivores	Richness	No	No
HerbRich	Herbivores	Richness	No	No
InvertRich	Invertivores	Richness	No	No
LenticRich	Lentic Species	Richness	No	No
OmnivoreRich	Omnivores	Richness	No	No
SensitiveRich	Sensitive Species	Richness	No	No
SGCNRich	Species of Greatest Conservation Need	Richness	No	No
SpeciesRich	Species Richness	Richness	Yes	No
TolerantRich	Tolerant Species	Richness	No	No
TopCarnRich	Top Carnivores	Richness	No	No

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each metric type were determined using a forward (add-one-in) and backward (leave-one-out) search using the "step" function in R. Top candidate models were ranked and compared based on Akaike information criterion (AIC).

Habitat score as a latent variable.—Results of initial modeling efforts indicated that it was not feasible to use fish community metrics to directly predict Topeka Shiner CPUE or presence/absence. Predictions generated by these initial models did not correlate with the known condition of sites within the test dataset. Therefore, we created a latent variable for habitat score and quantified the value of this unobserved but important environmental characteristic from the values of five observed variables: two variables positively associated with Topeka Shiner presence/absence (dissolved oxygen and turbidity; Simpson *et al.*, 2019), and three variables negatively associated with Topeka Shiner presence/absence (oxbow length, canopy cover, oxbow depth; Bakevich, 2012; Simpson *et al.*, 2019). Each of these five variables were measured on the same day the fish community was sampled and detailed methods for these measurements can be found in Bakevich *et al.*, (2012) and Simpson *et al.* (2019).We calculated each site's habitat score as follows: for each of the five variables, sites were grouped into quintiles (0–20th, 21st–40th, 41st–60th, 61st–80th, and 81st–100th), and scored based upon these quintiles (for positively correlated variables: 0–20th = 1, 21st–40th = 2, *etc.*; for negatively correlated variables: 0–20th = 5, 21st–40th = 4, *etc.*; Fig. 3). Scores were averaged across the five variables at each site, then, using the 0–10th, 11th–20th. . .91st–100th percentiles for these scores, sites were scored on a scale of 1–10.

Linear modelling with habitat score.—Additive linear models were fit to predict habitat score from the fish community metrics for each of the three metric types from the Topeka Shiner CPUE candidate metric set based on both a forward (add-one-in) and backward (leave-one-out) search using the "step" function in R. Top candidate models were compared using AIC. Predictors within top candidate models for each metric type were tested for significance by conducting an ANOVA based on the "joint tests" function in the package emmeans (Lenth, 2020). Generalized linear models were fit to predict Topeka Shiner CPUE (family "quassipoisson") and presence/absence (family "binomial") from habitat score.

To supplement model results—Modelling efforts detailed to this point were not successful in generating models that accurately predict Topeka Shiner presence/absence or abundance. Therefore, to further examine the feasibility of differentiating between oxbows in which Topeka Shiner are present versus those where they are absent, we performed nonmetric multidimensional (three dimensions) scaling (nMDS) on fish community metrics and environmental variables using a square-root transformation with the metaMDS function in the R package vegan using the Bray-Curtis distance measure (Oksanen *et al.*, 2020). Furthermore, we ran nMDS on all sites grouped by Topeka Shiner presence/absence. Finally, we used Topeka Shiner presence/absence as the explanatory variable in a distance-based redundancy analysis (dbRDA) in the package vegan. All analyses were conducted, and figures were generated with program R (version 3.6.3; R Core Team, 2020). P values were considered significant at $\alpha \leq 0.05$.

RESULTS

A total of 193,520 fish from 59 species were collected from the 137 sites included within the database. An average of 1412 ± 2390 (mean ± SD) individuals of 9.9 ± 4.7 species were collected at each site. Species richness values ranged from 1 to 21 and total abundance ranged from 1 to 13605 across all sites. Topeka Shiner were present at 57 of the 137 sites with a total of 3603 individuals collected. When present, Topeka Shiner catches ranged from 1 to 732 individuals with an average of 63.2 ± 129.6. Output of the PERMANOVA test indicated

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TABLE 2.—Top candidate models for predicting Topeka Shiner CPUE from fish community metrics

Metric type	Top candidate model	ΔAIC
Singlespecies	CreekChubCPUE FatheadMinnowCPUE OrangespottedSunfishCPUE	0.00
Richness	PiscivoreRich SpeciesRichness	20.19
Composition	CompGrOr	37.38

that sites where Topeka Shiner are present can be differentiated from sites where the species is absent (F-ratio: 3.067 on 1 and 135 d.f., $P = 0.006$), and that this difference is driven mostly by the abundance of Fathead Minnow (*Pimephales promelas*), Common Shiner (*Luxilus cornutus*), Orangespotted Sunfish (*Lepomis humilis*), White Sucker (*Catostomus commersonii*), and Brassy Minnow (*Hybognathus hankinsoni*).

TOPEKA SHINER AS RESPONSE, FISH COMMUNITY METRICS AS PREDICTOR

Generalized linear models for prediction of Topeka Shiner CPUE based on single species metrics indicated a top candidate model which included CreekChubCPUE, FatheadMinnowCPUE, and OrangespottedSunfishCPUE as predictors (metric definitions can be found in Table 1). Models for prediction of Topeka Shiner CPUE based on composition metrics indicated a top candidate model which included CompGrOr as the only predictor. Model results for richness metrics indicated a top candidate model which included PiscivoreRich and SpeciesRichness as predictors of Topeka Shiner CPUE. Comparison of AIC for the three top candidate models indicated the model based on single species metrics was the best supported model (Table 2). However, analysis of the top candidate model based on predicted Topeka Shiner CPUE from the test dataset indicated no association between predicted and actual CPUE (adjusted R-square: 0.07).

For Topeka Shiner presence/absence metrics, model results for single species metrics indicated a top candidate model which included CommonCarpCPUE, LargemouthBassCPUE, PiscivoreCrChCPUE, SGCNCPUE, and YellowBullheadCPUE as predictors of Topeka Shiner presence/absence. Model results for composition metrics indicated a top candidate model with only one predictor of Topeka Shiner presence/absence, CompGrOr. Model results for richness metrics indicated a top candidate model that included TopCarnRich and CentRich for predicting Topeka Shiner presence/absence. Comparison of AIC for the three top candidate models indicated the model based on single species metrics was the best supported model (Table 3). However, analysis of the top candidate model based on predicted Topeka Shiner presence/absence from the test dataset indicated no association between predicted and actual presence/absence (adjusted R-square: 0.02).

HABITAT SCORE AS A LATENT VARIABLE

Habitat score as response, fish community metrics as predictor.—Both the forward and backward selection techniques suggested the model SiteHabitatScore $\sim$ BlackBullhead-CPUE + WhiteSuckerCPUE as the top candidate model predicting SiteHabitatScore from single species metrics. However, ANOVA results indicated that neither species have a statistically significant effect on habitat score at the $P < 0.05$ level, although the P for

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TABLE 3.—Top candidate models for predicting Topeka Shiner presence/absence from fish community metrics

Metric type	Top candidate model	ΔAIC
Single species	CommonCarpCPUE, LargemouthBassCPUE, PiscivoreCrChCPUE YellowBullheadCPUE	0.00
Richness	TopCarnivoreRich CentRich	13.30
Composition	CompGrOr	13.92

BlackBullheadCPUE was approaching significance and for WhiteSuckerCPUE was also relatively low (F-ratio: 3.705 on 1 and 134 d.f., P = 0.056 for BlackBullheadCPUE; F-ratio: 4.194 on 1 and 134 d.f., P = 0.1206 for WhiteSuckerCPUE). For composition metrics the forward and backward searches suggested SiteHabitatScore ~ CompGrOr as the top model. ANOVA results indicated that CompGrOr has a significant effect on habitat score (F-ratio: 4.436 on 1 and 135 d.f., P = 0.037). Forward and backward searches for richness metrics indicated two top candidate models, each containing a single metric; SpeciesRich, and PiscivoreRich. Comparison of AIC for the four top candidate models indicated that the model based on composition metrics is the best supported model (Table 4). These results indicate that site habitat score can be best modelled by the composition metrics model containing CompGrOr.

Topeka shiner as response, habitat score as predictor.—Model results indicated that site habitat score is not a significant predictor of either Topeka Shiner presence/absence (F-ratio: 0.004 on 1 and Inf d.f., P = 0.9501), or Topeka Shiner CPUE (F-ratio: 2.758 on 1 and Inf d.f., P = 0.0968). Given the poor performance of models on the test dataset, multiple follow-up analyses were conducted to determine if the models could be improved. Non-metric multidimensional scaling (nMDS) on fish community metrics and environmental variables revealed no clear grouping of metrics or variables that are associated with Topeka Shiner (Stress = 0.129, Fig. 4). Next, we conducted nMDS on sites grouped by the presence or absence of Topeka Shiner, but analyses did not reveal any clear groupings (Stress = 0.167, Fig. 5). Lastly, dbRDA indicated that the presence or absence of Topeka Shiner accounted for only 7.019% of the variation in the fish community data. These findings support the conclusion that fish community data do not contain a sufficient signal to use fish community metrics as a predictor for Topeka Shiner CPUE or presence/absence, even with the introduction of habitat score as an "intermediate" latent variable.

TABLE 4.—Top candidate models for predicting site habitat score from fish community metrics

Metric type	Top candidate model	ΔAIC
Composition	CompGrOr	0.00
Single species	BlackBullheadCPUE WhiteSuckerCPUE	0.39
Richness	PiscivoreRichness	1.44
	SpeciesRichness	3.23

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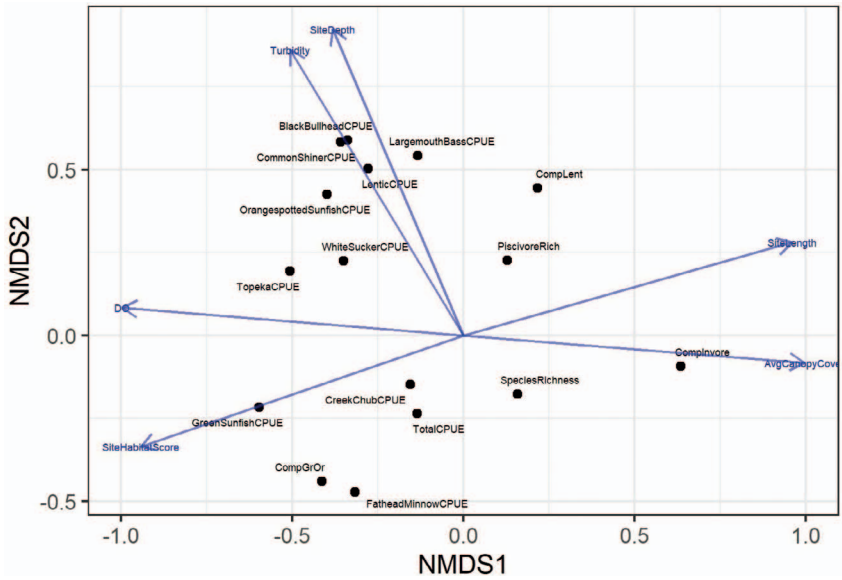


FIG. 4.—nMDS ordination of fish community metrics and environmental variables

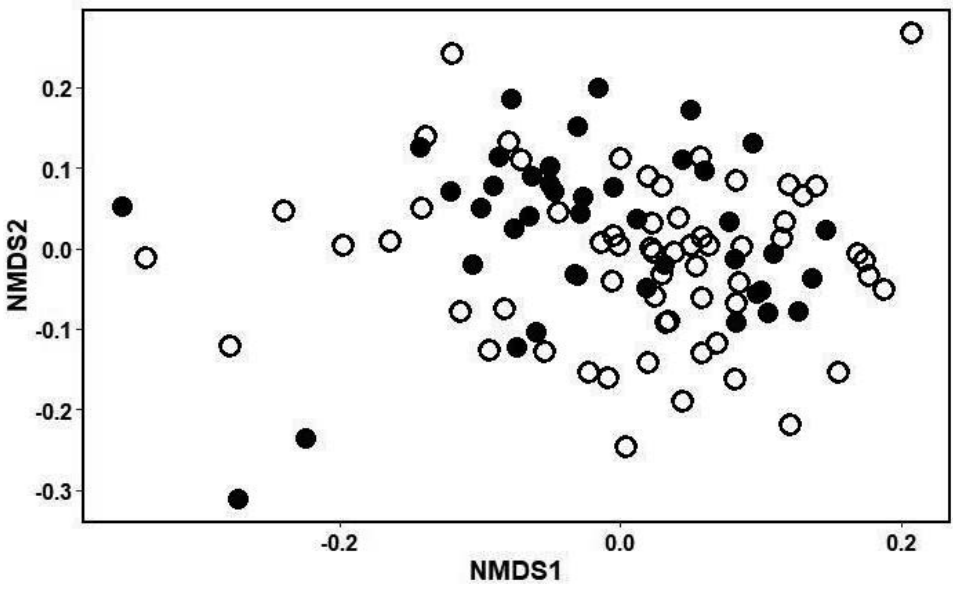


FIG. 5.—nMDS ordination of sites, identified by Topeka Shiner presence (black) or absence (white)

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TABLE 5.—Summary of Topeka Shiner associations with fish community metrics and abiotic factors as determined by previous studies

Study	Biotic or abiotic	Association with Topeka Shiner	Direction of association
Bakevich 2012	Biotic	Lentic Species CPUE	Positive
		Fathead Minnow CPUE	Positive
		Submerged Vegetation	Positive
	Abiotic	Oxbow Depth	Negative
		Canopy Cover	Negative
Simpson 2018	Abiotic	Oxbow Depth	Negative
		Canopy Cover	Negative
Simpson et al. 2019	Biotic	Brassy Minnow CPUE	Positive
		Orangespotted Sunfish CPUE	Positive
	Abiotic	Dissolved Oxygen	Positive
		Turbidity	Positive
		Oxbow Length	Negative
Thomson and Berry 2009	Abiotic	Oxbow Depth	Negative
Winston 2002	Biotic	Fathead Minnow Presence	Positive
		Black Bullhead Presence	Positive

DISCUSSION

Our findings indicate that Topeka Shiner CPUE or presence/absence within oxbows cannot be accurately predicted by the fish community present within an oxbow at the time of sampling. Therefore, at this time, an effective OQI based upon fish assemblage metrics cannot be created. Given the size of the database analyzed, we are confident that this finding reflects the weakness of relationships within oxbow fish communities and is not an artifact of small sample size. Additionally, while fish community metrics were not found to accurately predict Topeka Shiner CPUE or presence/absence by our study, multiple studies have found a set of biotic and abiotic habitat variables to be associated with Topeka Shiner CPUE or presence/absence and restoration practitioners should use these variables to inform designs of future oxbow restorations.

Although previous, individual oxbow studies found relationships between Topeka Shiner CPUE and presence/absence and fish community metrics, these relationships were not consistent across studies (Bakevich, Pierce, and Quist, 2013; Simpson *et al.*, 2019; Table 5). However, studies conducted within streams have found stronger relationships between Topeka Shiner CPUE and presence/absence with fish community metrics (Blausey, 2001; Gerkin and Paukert, 2013, Simpson *et al.,* 2021). Therefore, it is possible that our inability to effectively model Topeka Shiner CPUE or presence/absence with fish community metrics within oxbows is due to the sampling methodology for oxbow fish communities and the "island" like stochasticity of oxbow fish communities.

OXBOWS AS ISLANDS

During flood events, fishes swim freely throughout the inundated floodplain. As flood waters recede, fishes either return to the main stream channel or colonize floodplain aquatic habitats such as oxbows. Concurrently, oxbows become disconnected from the main stream channel and these habitats effectively become aquatic "islands" within the floodplain as fishes can no longer leave from, or enter into the oxbow. Dynamics of this process have been

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described in oxbows (Fischer *et al.*, 2018), and it is analogous to processes described by island biogeography theory (Macarthur and Wilson, 1967). If the process by which fish enter these oxbows is random, we can assume that no quantifiable relationships exist within the fish community at the time of isolation (colonization of oxbows as floodplain water levels are receding). However, as time progresses following isolation, we may begin to see patterns emerge within the fish community as the result of competitive or piscivorous interactions. For example, if an oxbow contains a high proportion of piscivores with the rest of the fish community made up of small cyprinids at the time of isolation from the stream, we may expect to find that as time progresses, the fish community will become more and more dominated by piscivores until either an equilibrium is reached, or the piscivores deplete all other species. In effect, relationships may begin to form within the fish community over time based on the fish community that was present at isolation, and these relationships may be useful in modelling Topeka Shiner presence/absence or CPUE. However, given the frequent restructuring of the fish community within an oxbow as the result of flooding (approximately once every 1.5 to 2 y; Schilling *et al.*, 2019), and asynchronous cycles of both flooding and drying (Fischer *et al.*, 2018), it is possible that the oxbow would not exist as an island long enough for a fish community to reach equilibrium. If an equilibrium is not reached before flooding restructures the fish community, the fish community within the oxbow would exist in a perpetually stochastic state, and we would expect to not be able to effectively predict Topeka Shiner presence/absence or CPUE within oxbows. Our findings, and the inability of previous studies to corroborate associations between Topeka Shiner and specific fish community metrics, point towards this as a possibility.

OXBOW VS. STREAM SAMPLING

The difference in success modelling Topeka Shiners between oxbows and streams may be attributed in part to the way in which fish community structure may become established within these respective habitat types, and how we sample these habitats. Within the stream channel, fishes are free to move about through various habitat types and associate with preferred habitat which leads to predictable fish assemblages within specific areas (example; more lentic species in pools than in riffles). Although fishes may be able to select for and associate with specific micro-habitats within oxbows, typical sampling of oxbow fish communities involves sampling the entire oxbow as one unit with no attempt to measure within-oxbow habitat differences. This sampling methodology would eliminate any habitat associations that may structure fish communities at the micro-scale within the oxbow. In contrast, stream sampling can be targeted to specific habitat types (pools, riffles, *etc.*), which allows for the fish communities associated with these specific habitat types to be distinguished. Even if fish community structure differs throughout an oxbow depending on habitat, these differences are likely homogenized during sampling. Therefore, even if Topeka Shiner associate with a specific fish community within the oxbow based upon micro-habitat (as they do within pools in streams), these associations are lost during sampling.

DEFINING SUCCESS AND RESTORATION NEXT STEPS

Given the apparent variability of fish communities within oxbows, it is likely that attempting to define a successful restoration at the oxbow level is not feasible. Instead, success should be defined more broadly as the increase in population and range of Topeka Shiner within basins where oxbows have been restored. There is indication that current restoration practices are being successful in recovering Topeka Shiner populations and

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expanding the range of the species in basins where oxbows are being restored. For the database we analyzed, Topeka Shiner occurred at similar proportions of restored and unrestored sites (41% of restored oxbows and 42% of unrestored oxbows). However, Topeka Shiner were found in higher abundances within restored oxbows than within unrestored sites (35.02 ± 108.32 and 11.12 ± 35.68 respectively; mean ± SD). Furthermore, Topeka Shiner catches during the 2019 and 2020 field season were consistently higher than those reported by previous studies in the same region (Bakevich, 2012; Bybel, 2018; Simpson, 2018). During the summer of 2020, our research group collected 3385 Topeka Shiner from twelve recently restored oxbows which included 1731 individuals from a single restored oxbow. Additionally, we documented the return of the Topeka Shiner to the White Fox Creek HUC10 of central Iowa, a basin in which Topeka Shiner had not been collected in 36 years despite multiple sampling efforts (Osterhaus *et al.*, 2021). Many oxbows have been restored within the White Fox Creek HUC10 over the last decade, providing high quality habitat for Topeka Shiner that dispersed into the basin.

Given that current restoration practices are apparently successful in increasing Topeka Shiner populations and range, we recommend that restoration efforts be targeted to areas of the range in which oxbow habitat is currently lacking. Additionally, while relationships between Topeka Shiner and fish community metrics have been weak and not reproducible, multiple abiotic metrics have been determined to be significant predictors of Topeka Shiner presence or absence by other studies (Table 5). However, for oxbows in the database we analyzed, Topeka Shiner occurrence did not vary with the environmental variables we tested (Fig. 6). While the differences were not significant, Topeka Shiner did occur more often in oxbows that were more turbid, had lower canopy cover, were shorter in length, and were smaller in surface area (Fig. 6). These findings corroborate those of multiple other studies (Table 5). Therefore, we recommend that restoration practitioners should use these metrics to inform the design of future restorations. Additionally, once oxbow habitat has been restored to match historical levels in basins from which Topeka Shiner have been extirpated, repatriation efforts may be needed to recover populations in these areas. Given that Topeka Shiner can be cultured in controlled systems (Katula, 2015; Campbell *et al.*, 2016; Bybel, 2018; U.S. Fish and Wildlife Service, 2021), repatriation efforts may play a critical role in the future recovery of the species in basins from which they were extirpated.

CONCLUSION

Our findings point to the random nature by which fishes find their way into oxbows during flood events. If this process were deterministic (*i.e.*, fishes choosing oxbows based upon habitat characteristics), we would expect to be able to differentiate between fish communities where Topeka Shiner are present versus those where Topeka Shiner are absent as underlying habitat preferences would structure the fish communities within the oxbows. Although creation of an OQI is not possible with the database and analyses we implemented, success of oxbow restorations can be determined more broadly by examining changes in Topeka Shiner populations and range in basins where oxbow restorations have occurred. Furthermore, multiple biotic and abiotic habitat factors have been determined to be significant predictors of Topeka Shiner presence/absence and abundance and restoration practitioners should use these factors to inform the design of future restorations.

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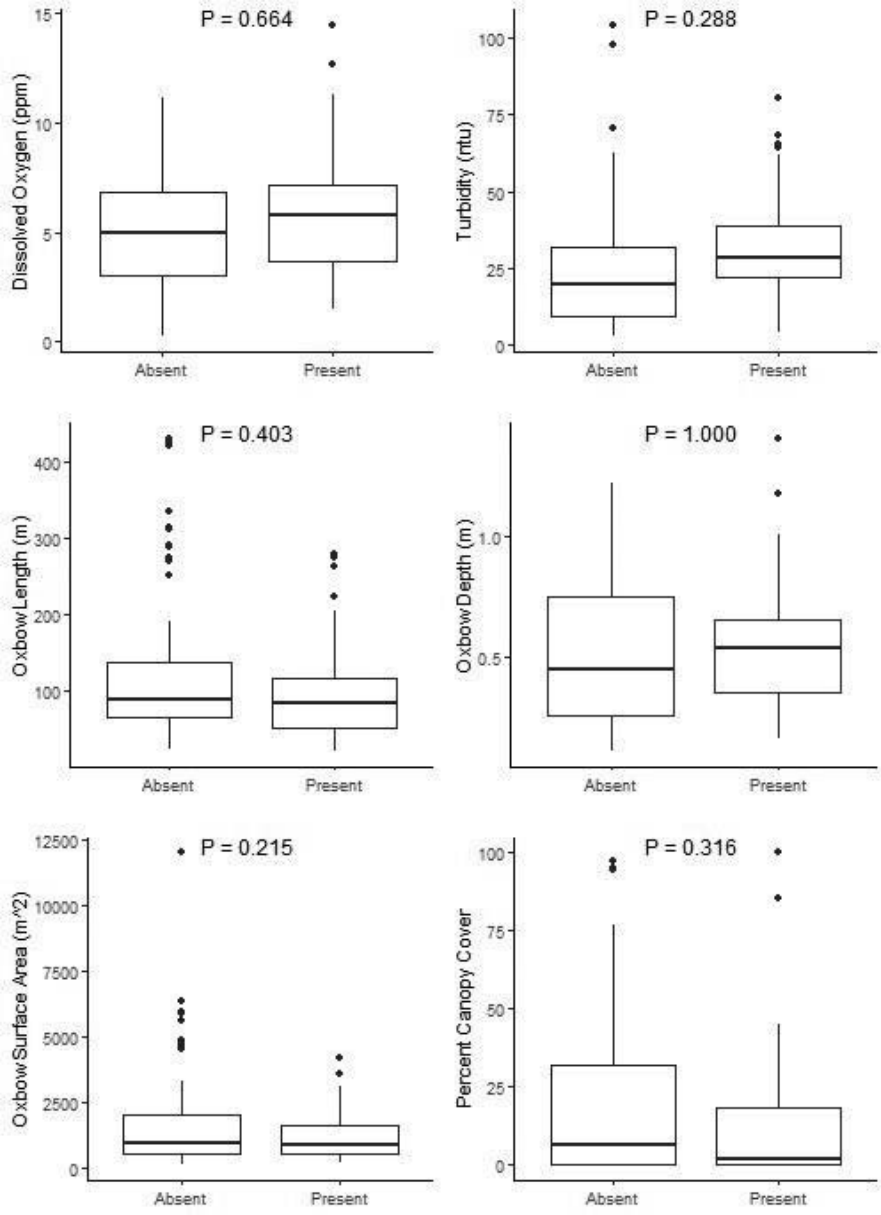


FIG. 6.—Abiotic variables for oxbows where Topeka Shiner were absent versus present for the database analyzed in this study (P-values from separate variance *t*-tests, adjusted for multiple comparisons using Bonferroni method)

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